CAPSTONE PROJECT PROPOSAL

University of Information Technology and Sciences (UITS)

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Smart Bangla AI Voice Assistant & IoT Controller for Mental Wellness and Ambient Automation [MAYA]

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Submitted by

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1. Introduction

Commercial voice assistants such as Apple Siri, Amazon Alexa, and Google Assistant have transformed human-computer interaction across the globe. However, these systems predominantly cater to English and other high-resource Western languages, offering negligible or severely limited comprehension of spoken **Bangla**—the native language of over 250 million people worldwide. In Bangladesh, this linguistic divide disproportionately isolates the elderly, individuals with low digital literacy, and those in non-urban communities.

Simultaneously, contemporary society faces a rapidly escalating mental health crisis characterized by widespread loneliness, stress, and clinical depression. In Bangladesh, mental healthcare remains burdened by social stigma, an acute shortage of licensed counselors, and financial barriers. Many struggling individuals find it difficult to articulate their emotional pain to family members or seek professional psychological help until acute crisis points occur.

This project proposes **MAYA**, an affordable, low-latency, IoT-enabled intelligent voice assistant designed specifically for Bangla speakers. Beyond answering general queries and controlling home appliances, MAYA incorporates an **empathetic companion and depression-rescue engine**. Built on an Edge–Gateway hybrid architecture utilizing the ESP32-S3 microcontroller and a modular Python backend, MAYA provides non-judgmental active listening, emotional de-escalation, distress detection, and immediate referral to verified emergency crisis hotlines (such as *Kaan Pete Roi*), while simultaneously delivering voice-driven smart-home appliance control.

2. Problem Statement

Most existing commercial voice systems fail in the context of everyday Bangladeshi life due to several fundamental shortcomings:

- **Language & Dialect Gap:** Mainstream assistants fail to parse standard conversational Bangla, regional phonetics, and colloquial code-mixed speech ("Banglish").
- **Lack of Emotional Intelligence:** Conventional voice assistants operate as cold transactional command parsers; they cannot recognize sadness, chronic loneliness, or severe emotional distress.
- **Mental Health Stigma and Accessibility:** Seeking therapy in Bangladesh carries social stigma and high monetary costs. Accessible, non-judgmental first-line emotional support is practically non-existent.
- **Hardware Cost & Complexity:** Implementing edge AI on expensive single-board computers (such as Raspberry Pi 5) escalates product costs beyond what ordinary households can afford.
- **Smart Home Inaccessibility:** Modern smart-home setups require complex proprietary applications in English, excluding non-technical and non-English-speaking users from ambient home control.

3. Objectives

General Objective

To design, implement, and evaluate an affordable, IoT-based Bangla AI voice assistant (**MAYA**) that combines natural conversational capabilities in spoken Bangla, empathetic mental health support and depression de-escalation, and hands-free smart-home automation.

Specific Objectives

- **Edge Hardware Interface:** Develop low-latency I²S digital audio capture and playback firmware on an affordable **ESP32-S3 N16R8** microcontroller using the INMP441 MEMS microphone and MAX98357A Class-D amplifier.
- **Modular Streaming Gateway:** Engineer a robust Python (FastAPI) WebSocket gateway to coordinate real-time bidirectional audio streaming between edge hardware and cloud AI microservices.
- **Bangla Speech Pipeline:** Integrate and optimize low-latency Speech-to-Text (STT) and neural Text-to-Speech (TTS) models tailored for standard Bangladeshi Bangla (`bn-BD`).
- **Empathetic Companion & Depression Rescue Engine:** Formulate Cognitive Behavioral Therapy (CBT)-inspired conversational logic and a deterministic safety triage layer that detects crisis keywords, delivers supportive de-escalation, and provides emergency hotline contacts.
- **IoT Smart-Home Automation:** Implement native LLM tool/function calling to trigger physical relays for controlling household loads (e.g., lighting and fans).
- **Empirical Evaluation:** Benchmark speech recognition Word Error Rate (WER), emotional classification accuracy, end-to-end response latency, and IoT command reliability across diverse Bangla speakers.

4. Scope and Limitations

In Scope

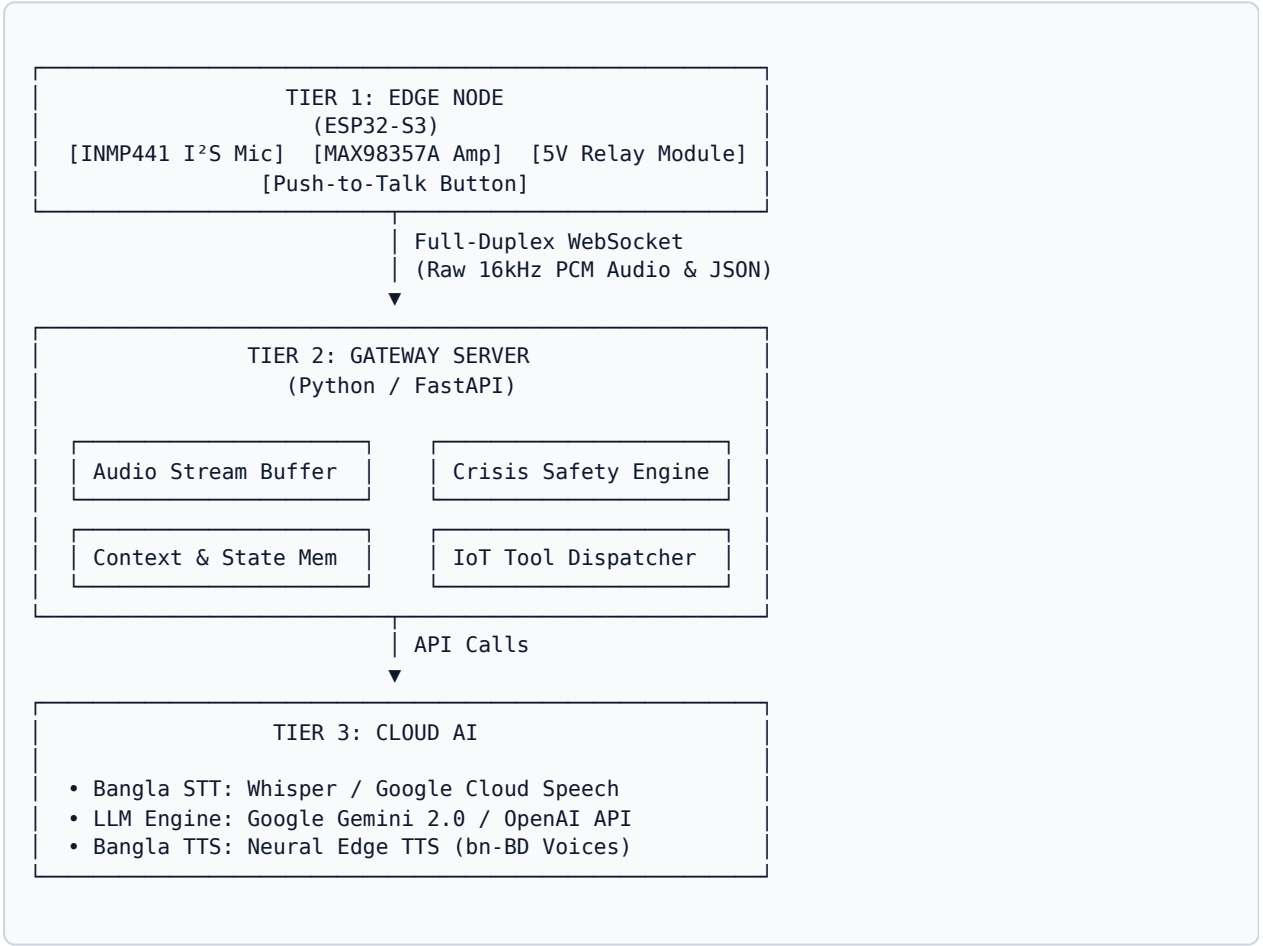
- Speech-to-text transcription in standard conversational Bangla and common mixed Bangla-English phrasing.
- Conversational reasoning across general topics (weather, time, explanations, daily conversations).
- Empathetic listening, sentiment analysis, and safety-screened suicide/depression crisis de-escalation.
- IoT appliance control through a 1-channel optocoupled relay module.
- Working physical prototype using the ESP32-S3, digital I²S audio hardware, and a local/cloud FastAPI companion server.

Limitations & Out of Scope

- **Not a Medical Device:** MAYA serves as an empathetic companion and crisis triage tool; it does not diagnose psychiatric disorders or replace licensed clinical psychotherapy.
- **Network Requirement:** While edge appliance control operates locally, natural language understanding and neural speech synthesis require an active internet connection.
- **Regional Dialects:** The prototype targets standard spoken Bangla; highly diverged regional dialects (e.g., Chittagonian, Sylheti) are outside the primary baseline.

5. System Architecture

MAYA adopts a **3-Tier Edge–Gateway–Cloud Hybrid Architecture** that decouples physical hardware I/O from heavy machine learning workloads:



6. Key System Modules

Module Name	Environment	Core Functionality
Edge Audio & I/O	ESP32-S3 (C++ / Arduino)	Samples I²S audio at 16kHz, handles Wi-Fi socket streaming, outputs digital audio to amplifier, and actuates relay GPIOs.
Gateway Router	FastAPI / Python	Manages sessions, buffers incoming audio, controls turn-taking, and coordinates downstream AI services.
Bangla Speech Pipeline	Whisper & Edge-TTS	Converts spoken Bangla audio to text and synthesizes empathetic responses back into natural Bengali voice (`bn-BD`).
Depression & Crisis Triage	Python Regex & Classifier	Screens transcribed text for high-risk emotional markers. Enforces priority intervention protocols when self-harm is detected.
Empathetic Reasoning	Gemini / GPT Prompt Guard	Conducts conversational dialogue with warm, CBT-aligned active listening; invokes function calls for appliance commands.
IoT Actuator	Python Gateway + ESP32	Dispatches validated JSON hardware payloads (`{"cmd": "RELAY", "val": 0|1}`) to switch appliances.

7. Hardware & Software Specifications

A. Hardware Components

Component	Specification & Function	Est. Cost (BDT)
ESP32-S3 N16R8	Dual-core Xtensa LX7, 16MB Flash, 8MB PSRAM, Wi-Fi 4 + BLE 5	৳800–1,000
INMP441 Microphone	Omnidirectional digital MEMS microphone with I²S interface	৳200–330
MAX98357A Amplifier	3.2W Class-D mono digital-to-analog amplifier breakout (I²S)	৳180–300
4 Ω, 3 W Speaker	Mini acoustic transducer for clear Bangla voice playback	৳100–200
5 V Relay Module	1-Channel optocoupled relay for appliance/lighting control	৳80–150
Power Supply & Wiring	Regulated 5V 3A adapter, breadboard, tactile button, jumper wires	৳450–700

B. Software Tools & Libraries

Layer	Technologies	Purpose
Embedded Firmware	C++, Arduino Framework, FreeRTOS, ArduinoJson	Hardware I/O, I²S audio buffering, WebSocket client
Gateway Server	Python 3.10+, FastAPI, Uvicorn, WebSockets	Asynchronous session routing & pipeline coordination
Speech & AI Models	OpenAI Whisper, Google Gemini 2.0 Flash, Edge-TTS	Bangla STT, empathetic reasoning, neural Bangla voice

8. Methodology and Project Timeline

The project follows an iterative 12-week schedule with measurable engineering milestones:

Weeks	Phase	Tasks & Deliverables
1–2	Literature & Design	Finalize architecture, assemble BOM, establish evaluation criteria, design API schemas.
3–4	Hardware Audio Validation	Wire ESP32-S3 to INMP441 and MAX98357A. Verify bidirectional I²S capture and playback.
5–6	Gateway & Speech Pipeline	Implement FastAPI WebSocket server. Integrate Bangla STT (Whisper) and Bangla TTS (Edge-TTS).
7–8	Empathetic AI & Crisis Engine	Develop CBT empathetic system prompts, emotion classification layer, and crisis hotline safety triggers.
9–10	IoT Integration & Actuation	Implement LLM tool calling and relay switching. Run integration tests on end-to-end user loops.
11	Evaluation & Benchmarking	Measure WER, sentiment classification accuracy, IoT success rate, and latency with multiple speakers.
12	Final Report & Defense	Compile thesis document, prepare demonstration rig, and conduct Capstone presentation.

9. Testing and Evaluation Metrics

Evaluation Metric	Target Benchmark	Verification Method
STT Word Error Rate (WER)	≤ 18%	Tested against 100 recorded Bangla speech samples across diverse accents.
Crisis Detection Recall	≥ 95%	Tested against suicidal and acute emotional distress trigger datasets.
IoT Command Success Rate	≥ 95%	Ratio of successfully executed relay toggles over 100 spoken smart-home commands.
End-to-End Latency	≤ 2.5 seconds	Time measured from speech completion to initial speaker audio output.
User Empathy Rating	≥ 4.2 / 5.0	Subjective Likert-scale feedback evaluated by test participants assessing conversational warmth.

10. Risks, Ethics, and Mitigation Strategies

- **Mental Health Safety:** Erroneous advice or failure to recognize suicide risk is strictly mitigated through hardcoded deterministic keyword guardrails that override the LLM whenever crisis phrases are uttered, immediately providing registered emergency contacts (**Kaan Pete Roi: 01779-554391**).
- **Acoustic Feedback & Echo:** Prevented by using a tactile push-to-talk mechanism and muting the microphone during active speaker playback.
- **Network Latency:** Minimized through asynchronous streaming WebSockets and fast inference APIs (Groq Whisper & Gemini Flash).
- **Data Privacy:** No audio is captured or streamed until the physical push-to-talk button is deliberately held by the user.

11. Expected Outcomes

- A fully functional physical prototype of **MAYA** built with the ESP32-S3 and breadboard peripherals.
- A low-latency conversational AI pipeline capable of empathetic dialogue in natural spoken Bangla.
- A robust safety triage mechanism providing active depression and emotional crisis de-escalation.
- Voice-controlled physical relay actuation for domestic smart-home loads.
- Comprehensive thesis report, dataset benchmarks, and reproducible open-source documentation.

12. Conclusion

MAYA bridges a critical intersection between embedded IoT systems, applied artificial intelligence, and public mental wellness in Bangladesh. By combining affordable microcontroller hardware with state-of-the-art conversational and empathetic speech intelligence, this project demonstrates that low-cost, high-impact technology can bring accessible voice automation and compassionate support to millions of native Bangla speakers.